AMITY UNIVERSITY KOLKATA

Weekly Progress Report

For the Week Commencing 17 August 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Android route selection. Added explicit Home-to-Office and Office-to-Home options before recording a commute.

Achievements for the week

Connected the selected direction to session metadata so the two routes can be separated during later processing.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	17 Aug: Built the two route choices. The direction is now available before a recording session begins.
Tuesday	18 Aug: Stored the selected direction with each recording. The saved metadata identifies which route was selected.
Wednesday	19 Aug: Tested starting and stopping both route types. Both choices were exercised with separate test recordings.
Thursday	20 Aug: Checked that morning and evening sessions remained distinguishable. The route value remains available when the session is exported.
Friday	21 Aug: Recorded the route-selection behavior for the Android workflow. The recording flow now supports the two commute directions consistently.

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Weekly Progress Report

For the Week Commencing 24 August 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Five-second GPS polyline updates. Added a location update loop that pulls a GPS point every five seconds.

Achievements for the week

Drew the recorded path as a translucent polyline so movement is visible while the session is running.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	24 Aug: Added the five-second location interval. The timer controls the interval between consecutive location requests.
Tuesday	25 Aug: Created the live translucent polyline. The line is updated without hiding the underlying map.
Wednesday	26 Aug: Updated the line as new points arrived. New points are appended in chronological order.
Thursday	27 Aug: Checked the line during both commute directions. The trace was observed while the route direction changed.
Friday	28 Aug: Verified that the displayed path matched the saved points. The live display and exported coordinates describe the same path.

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Weekly Progress Report

For the Week Commencing 31 August 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Satellite status and signal display. Added the number of visible satellites and the number reporting a usable fix.

Achievements for the week

Displayed satellite signal strength values and used 25 as a practical indication of good strength without assigning an unsupported unit.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	31 Aug: Read the satellite status values. The status panel now distinguishes visibility from a usable fix.
Tuesday	01 Sep: Separated visible satellites from satellites with a fix. The fix count changes independently from the total satellite count.
Wednesday	02 Sep: Displayed the signal-strength readings. The readings are shown without guessing a measurement unit.
Thursday	03 Sep: Tested the status while the phone moved outside. The test confirmed that open-sky conditions improve the readings.
Friday	04 Sep: Recorded how satellite availability affected the recording. The information gives the user a direct indication of GPS quality.

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Weekly Progress Report

For the Week Commencing 07 September 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: On-device session database. Created the Android phone database for storing sessions and their location points.

Achievements for the week

Added deletion of old sessions from the phone so accumulated recordings do not keep slowing the application.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	07 Sep: Created the session tables and relationships. The schema keeps each session separate from its location-point records.
Tuesday	08 Sep: Saved location points under the active session. Points are associated with the correct active session.
Wednesday	09 Sep: Loaded the stored sessions for review. The session list can be read back after recording stops.
Thursday	10 Sep: Deleted an old session directly from the phone. Deletion removes the selected session instead of only hiding it.
Friday	11 Sep: Checked that deleting a session reduced the stored backlog. The phone no longer needs to retain every old recording indefinitely.

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Weekly Progress Report

For the Week Commencing 14 September 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Leaflet map marker and caching. Created the GPS locator marker with Leaflet and integrated the map display with the recorded route.

Achievements for the week

Followed Leaflet's caching-related implementation so the map and marker remain usable during repeated route viewing.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	14 Sep: Added the Leaflet map and GPS marker. The marker follows the latest GPS coordinate on the map.
Tuesday	15 Sep: Updated the marker as the current position changed. The map position and current route direction remain visible together.
Wednesday	16 Sep: Drew the route behind the marker. The path remains readable while the marker moves forward.
Thursday	17 Sep: Checked cached map behavior during repeated use. Repeated map loads use the configured cached resources.
Friday	18 Sep: Verified that the locator and polyline stayed synchronized. The map layer, marker, and route line now form one usable view.

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Weekly Progress Report

For the Week Commencing 21 September 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Android diagnostics and obstruction alert. Added a debug log for application errors and recording failures.

Achievements for the week

Added a 30-second vibration alert to tell the user to move the phone outside when satellite communication is obstructed.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	21 Sep: Added structured debug messages for failures. The log records enough context to identify where a failure occurred.
Tuesday	22 Sep: Triggered the obstruction timer after 30 seconds. The timer is tied to the period without a usable satellite update.
Wednesday	23 Sep: Added phone vibration feedback. The phone gives an immediate physical warning instead of relying only on the screen.
Thursday	24 Sep: Tested the alert with poor satellite visibility. The test reproduced the warning under obstructed conditions.
Friday	25 Sep: Checked that the alert helped identify blocked GPS conditions. The alert directs the user to move the phone outside before retrying.

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Weekly Progress Report

For the Week Commencing 28 September 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Permissions and battery behavior. Added the required location permissions and handled the phone's battery restrictions for recording.

Achievements for the week

Configured the relevant battery settings so the Android process can continue collecting GPS points without being stopped silently.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	28 Sep: Requested the required location permissions. The application requests access before starting location collection.
Tuesday	29 Sep: Handled permission denial and re-entry. The recorder returns to a usable state after the permission flow.
Wednesday	30 Sep: Reviewed battery optimization settings. Battery restrictions were reviewed as a separate cause of interrupted tracking.
Thursday	01 Oct: Tested recording while the screen and app state changed. The recorder was checked during background and screen-off conditions.
Friday	02 Oct: Verified that the recorder continued collecting points. Location collection is now protected against the main permission and battery interruptions.

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Weekly Progress Report

For the Week Commencing 05 October 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Browser pruning and eraser. Built the browser pruning workflow with an eraser for removing bad GPS observations.

Achievements for the week

Used the eraser to remove the least reliable points from the stack of displayed polylines and reconnect the remaining path.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	05 Oct: Loaded multiple recorded polylines. The eraser operates on the visible route data rather than deleting the source files blindly.
Tuesday	06 Oct: Selected noisy and off-road points. Neighboring points are joined after the selected point is removed.
Wednesday	07 Oct: Removed points with the eraser. The cleaned line was compared with the original noisy trace.
Thursday	08 Oct: Checked the reconnection between neighboring points. The remaining path was checked for gaps caused by over-pruning.
Friday	09 Oct: Saved the cleaned trajectory for later processing. The edited trajectory can now be used as a cleaner input for snapping.

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Weekly Progress Report

For the Week Commencing 12 October 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: Route snapping and reference polylines. Implemented snapping of recorded points to representative morning and evening route polylines.

Achievements for the week

Calculated nearest-route positions and off-route distances so noisy coordinates can be compared on a common route.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	12 Oct: Selected reference sessions for both directions. The reference route provides a common geometry for both commute directions.
Tuesday	13 Oct: Projected GPS points onto the reference polylines. Each point receives a nearest position on that geometry.
Wednesday	14 Oct: Calculated along-route distance. The distance measure identifies observations that drift away from the route.
Thursday	15 Oct: Measured off-route error. The projected progress can be compared across different sessions.
Friday	16 Oct: Reviewed the snapped CSV outputs. The generated snapped files preserve the original session identity.

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Weekly Progress Report

For the Week Commencing 26 October 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: ETA features and route profiles. Built progress, speed, route-bin, and congestion features from the cleaned and snapped sessions.

Achievements for the week

Created route profiles with average speed, session support, coordinates, and congestion values for ETA prediction.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	26 Oct: Calculated monotonic route progress. The feature table combines route position with observed movement conditions.
Tuesday	27 Oct: Derived speed from progress and timestamps. Speed is calculated from consecutive progress and timestamp values.
Wednesday	28 Oct: Assigned 120-metre route bins. The bins make sessions comparable at fixed route distances.
Thursday	29 Oct: Aggregated profile speeds and support counts. Support counts show how much historical data backs each profile value.
Friday	30 Oct: Inspected congestion and choke-point candidates. The resulting profiles provide the inputs needed by the ETA models.

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Weekly Progress Report

For the Week Commencing 02 November 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: ETA model training. Trained Random Forest with 300 trees, HistGradientBoosting with depth 8, learning rate 0.05, and 500 iterations.

Achievements for the week

Compared XGBoost with 200 estimators, depth 8, learning rate 0.04, 0.85 row and column subsampling, using GroupShuffleSplit with a 25% test set.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	02 Nov: Prepared the eight ETA feature columns. The training table uses route type, start minute, weekday, progress, off-route distance, speed, profile speed, and support.
Tuesday	03 Nov: Trained the Random Forest baseline. The group split keeps rows from the same session out of both partitions.
Wednesday	04 Nov: Trained HistGradientBoosting and XGBoost. The three regressors were trained against the remaining-time target.
Thursday	05 Nov: Measured MAE, RMSE, R2, and weighted metrics. The metrics expose both ordinary and sample-weighted model behavior.
Friday	06 Nov: Saved model comparisons and learning diagnostics. The saved diagnostics make the model comparison reproducible.

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Weekly Progress Report

For the Week Commencing 16 November 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: ETA prediction and final integration. Implemented CLI ETA prediction using route, start time, progress, coordinates, destination, and optional session ID.

Achievements for the week

Blended the trained model with local route-profile ETA, session statistics, and upcoming choke points for the final prediction output.

Future work plans

Finalize the report evidence, package the generated outputs, and prepare the submission.

Week's Summary

Days / Time	
Monday	16 Nov: Tested morning and evening prediction inputs. The command accepts either progress values or coordinates near the reference route.
Tuesday	17 Nov: Checked latitude/longitude route-position lookup. The nearest profile bin supplies the local speed and support context.
Wednesday	18 Nov: Tested destination-aware ETA scaling. The destination path is scaled separately from the full route estimate.
Thursday	19 Nov: Verified session-aware blending and choke-point output. The output reports the model, local ETA, session blend, and choke points.
Friday	20 Nov: Completed the final report evidence. Packaged the generated outputs and submission files.